

Knowledge-Graph Completion, Structural Inference, and Recursive Compression

When a Knowledge System Fills Its Gaps With More of What It Already Believes

Knowledge Systems and Reality Drift - Note #5 v1

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The Basic Pattern

Knowledge graphs are incomplete. Sources omit relationships, extraction systems miss facts, entities remain disconnected, and long-tail knowledge is sparsely represented. Knowledge-graph completion attempts to infer missing edges from the graph's existing entities, relations, and structural patterns.

This can recover useful knowledge. If similar entities repeatedly participate in similar relations, a model can identify plausible connections that were never explicitly stored. Completion makes the graph easier to search and reason over.

But the model does not inspect the world directly. It learns from the graph's representation of the world. When that representation contains systematic omissions, popularity bias, extraction artifacts, or inherited classifications, completion can reproduce those patterns as new knowledge.

When Prediction Becomes Knowledge

Link-prediction systems assign scores to candidate triples: subject, relation, object. A high score indicates that the relation fits patterns learned from the graph. It does not by itself establish that the relation is true outside the graph.

The distinction can weaken in use. Predicted links improve retrieval, populate downstream systems, or guide further data collection. Once displayed beside observed relations, inferred edges may lose their epistemic status. Structural plausibility becomes representational fact.

The graph then begins filling its gaps with compressed regularities derived from itself. What is well represented becomes easier to infer. What is absent because of historical neglect remains unlikely because the structure contains little evidence from which to predict it.

How Recursive Completion Emerges

Stage 1 - Incomplete Graph

Observed and extracted facts form a partial representation of the domain.

Stage 2 - Pattern Learning

A model compresses recurring relational structures into rules, embeddings, or learned parameters.

Stage 3 - Link Prediction

Missing relations are scored according to their compatibility with the existing graph.

Stage 4 - Operational Reuse

Predictions enter search, recommendations, analytics, or later graph updates.

Stage 5 - Recursive Compression

Inferences derived from the representation become inputs to later representations, strengthening the patterns that produced them.

Recursive Compression and Epistemic Status

Recursive Compression occurs when outputs produced from compressed representations are reused as if they were fresh observations. Each cycle can increase coherence because new information conforms to patterns already present. At the same time, independent contact with the domain can weaken.

The critical safeguard is epistemic separation. Observed, extracted, asserted, inferred, and model-generated edges should not become indistinguishable. Provenance must identify not only a source but the transformation through which each claim entered the graph.

Completion should also be evaluated against external observation, not only held-out edges from the same graph. A model can predict the graph accurately while preserving the graph's blind spots.

The Representation Stack

A source statement becomes an extracted triple. The triple becomes part of a graph. The graph becomes a training surface for a completion model. A predicted relation becomes a retrieval path, summary, or generated answer.

At every level, the output can look more connected than the source evidence warrants. The graph's increasing density may reflect recovered reality, recursive inference, or both. Without provenance, those possibilities collapse into one representation.

Examples Across Systems

Scientific Knowledge: Dense relations among well-studied entities are easier to predict than relations involving neglected diseases, regions, or populations.

Commercial Graphs: Popular products and users accumulate more connections, making them easier to recommend and still more central.

Security Analysis: Inferred affiliations can help identify hidden networks but may also turn structural similarity into unsupported suspicion.

Generative AI: Predicted graph relations can enter retrieval context and be rendered as fluent factual answers without visible distinction from sourced claims.

Conclusion

Knowledge-graph completion addresses a real problem: useful knowledge is often missing from explicit records. The representational risk begins when structural compatibility is treated as equivalent to external truth.

A completed graph is not simply a fuller map. It is a map partly extended by its own internal regularities. It remains grounded only when inferred relations preserve their status and retain a path back to independent evidence.

Related Phrases and Concepts

- knowledge-graph completion
- link prediction
- relation prediction

- graph embeddings
- structural inference
- incomplete knowledge graph
- coverage bias
- recursive compression

Semantic Fidelity Project Resources

- [Research Library \(GitHub\)](#)
- [Articles & Essays \(Substack\)](#)
- [Semantic Fidelity Glossary](#)